As mentioned, there is a long list of properties that are equivalent to an $n \times n$ matrix being invertible. One of those that we did is that a matrix is invertible exactly when it has precisely n pivots. Another couple are:

(2) $Ax = 0$ has only the trivial solution
(3) The columns of A span $\mathbb{R}^n$

Why is (2) equivalent? Well in one direction if A is invertible then left multiplying $Ax = 0$ by its inverse gives

$$A^{-1}Ax = A^{-1}x$$

---->

$$x = 0$$

proving $Ax = 0$ does have only the trivial solution. And the other way, if $Ax = 0$ has only the trivial solution then every column of A is a pivot column, meaning A has n pivots which we showed last time implies invertible.

For (3) if columns of A span $\mathbb{R}^n$ then again every row of A has a pivot implying invertible and likewise invertible implies n pivots (so every row has one), meaning the columns do span $\mathbb{R}^n$.

Lot of those equivalences fall into place easily once we have a few of them.

Folks often ask if they should memorize all the statements in the <u>Invertible Matrix Theorem</u>, which is where the three mentioned statements come from.

No, that isn't the point. It's about understanding, seeing that different contexts can connect, so understanding more helps you gain a better grasp of things.

Good example is a real matrix A (whether square or not) which has a nonpivot column has linearly dependent columns and also has Ax = 0 having a free variable.

So this leads people to often saying "linearly dependent means there is a free variable". And that's not really the most appropriate statement as the question of whether a set of vectors is linearly independent is a different context than asking if a linear system has infinite solutions due to having a free variable.

But while that might not a great way to phrase things, they are realizing that both those issues can be resolved via pivots, they are connecting two different ideas with the help of that, which is good!

That's what Invertible Matrix Theorem is about, connecting/understanding different ideas and being able to draw different conclusions.

An <u>invertible linear transformation</u> is linear $S: \mathbb{R}^n \dashrightarrow \mathbb{R}^n$ for which there exists $S^{-1}: \mathbb{R}^n \dashrightarrow \mathbb{R}^n$ satisfying

$$S \circ S^{-1} = S^{-1} \circ S = \text{identity mapping}$$

Not much to say about this as already used identity mapping to motivate identity matrix and then used inverse function to motivate idea of inverse matrix.

Book more motivates it in opposite way, which is perfectly fine, just a different way of looking at it.

Interesting question: Say $S: \mathbb{R}^n \dashrightarrow \mathbb{R}^n$ is a one-to-one linear transformation (since square matrix once it's one-to-one it's onto as well).

Since one-to-one, this does have an inverse function. But is that inverse function a linear transformation?

Might seem obvious but shouldn't make hasty assumptions. After all on $\mathbb{R}$ the functions e^x and $\sin x$ (on appropriate restricted domain) have inverses which are totally different style from the original functions.

So why would inverse of a linear transformation have to be a linear transformation given that? Again should be careful about jumping to conclusions.

Looking at these as matrices is a good way to clarify how this all works.

The association can be useful for instance in:

<u>Ex:</u> $T(x_1, x_2, x_3) = (5x_1 - x_2, 3x_1 + x_3, x_1 + x_2 + x_3)$

Does T have an inverse? Well the associated matrix for T, which can be found via coefficients or plugging in the standard vectors, is

$$\begin{bmatrix} 5 & -1 & 0 \\ 3 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

So simply check if this matrix is invertible. That's one reason why that association is useful. And if it is then the inverse of the matrix will have associated linear transformation equal to T^{-1}, proving that the inverse of such a linear transformation (should it exist) is indeed a linear transformation.

2.3.5 Is the following matrix invertible?

$$\begin{bmatrix} 0 & 3 & -5 \\ 1 & 0 & 2 \\ -4 & -9 & 7 \end{bmatrix}$$

One convenient way to check: are columns linearly independent? Since no column is a multiple of another that could only happen if the third column is a linear combination of first two, i.e. are there constants c_1, c_2 for which

$$c_1 \begin{bmatrix} 0 \\ 1 \\ -4 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 0 \\ -9 \end{bmatrix} = \begin{bmatrix} -5 \\ 2 \\ 7 \end{bmatrix}$$

To make second row work only chance is $c_1 = 2$ and first row only chance is $c_2 = \frac{-5}{3}$

So does third row work with those?

$$2(-4) + \frac{-5}{3}(9) = -8 + 15 = -7$$

So third row does match, making columns of this linearly dependent so matrix is not invertible.

This worked nicely since the 0s made it convenient to check this. If entries had been a bit different maybe something else would have been better?

That's one big reason to have all those statements which are equivalent to matrix being invertible. Depending on the specific entries sometimes one condition might be easier to check than another.

For instance if the matrix is in REF or close, then it very likely would be convenient to check if it's invertible by checking it has the right number of pivots. But if it's very far from being in REF, then quite likely something else might be more convenient.

But it's not only that, just connecting all those ideas in general gives you a better understanding of the nature of things.

Be careful too. One thing alluded to is that such matrices if they have a pivot in every column, they also do in every row. And while this is true in square matrices, definitely not true for general matrices!

So be careful not to apply such ideas in contexts where they are not valid!

Chapter 3: Determinants

For 2 x 2 matrices determinant can be viewed, at least in part, as a convenient way to check if matrix is invertible without actually finding the inverse.

But ok that doesn't seem like a big deal as inverse of a 2 x 2 matrix is not difficult to find. However, as you saw 3 x 3 and beyond is much more work. Is there a convenient way to check invertible without going through full process of finding inverse for larger square matrices?

We discussed this a bit at end of last chapter, all those conditions which are equivalent to invertible. But determinant is the most consistent and usually easiest way to conveniently check that, just need to define it for larger square matrices.

3.1 Introduction

First let us assess an upper triangular matrix. Same way we turned systems with same number of equations as variables into upper triangular to solve them more easily, let us first assess when an upper triangular matrix is invertible or not.

An upper triangular matrix is invertible exactly when every diagonal entry is nonzero.

Why? If any diagonal entry is 0, columns can't span $\mathbb{R}^n$ (proved in worksheet) meaning matrix is not invertible (from last chapter). And if all diagonal entries nonzero then those are pivot positions, guaranteeing the full n pivots, making the matrix invertible (again last chapter).

So even if matrix is 100 x 100, if upper triangular, it's very easy to tell if it's invertible or not.

So because of this, converting a matrix to be upper triangular via row operations to check whether or not it's invertible seems a reasonable idea.

We haven't formally defined determinant yet so it's not clear how these row operations might affect that. But they do not affect invertibility for sure which is main focus for now so don't have to worry about that at moment. Take generic 3 x 3:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Assume $a_{11} \neq 0$ (can interchange rows if needed and if entire first column is 0, then definitely not invertible so nothing further needed then anyway).

Then turn first column into 0s aside from top entry as usual.

$$\sim \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} \cdot a_{11} & a_{22} \cdot a_{11} & a_{23} \cdot a_{11} \\ a_{31} \cdot a_{11} & a_{32} \cdot a_{11} & a_{33} \cdot a_{11} \end{bmatrix}$$

Via scaling later two rows. Now do each minus appropriate multiple of first row to give:

$$\sim \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22}\, a_{11} - a_{21}\, a_{12} & a_{23}\, a_{11} - a_{21}\, a_{13} \\ 0 & a_{32}\, a_{11} - a_{31}\, a_{12} & a_{33}\, a_{11} - a_{31}\, a_{13} \end{bmatrix}$$

How we proceed here depends on whether the blue and/or red entry is 0. But ok this is just to get an idea, not being super specific so just assume blue is nonzero.

Under that assumption, make red entry 0 if not already so.

Once done, then the new value of the green entry determines invertibility as it is then the only potential zero entry on the diagonal of a matrix which is now upper triangular.

So that new green entry associates with the <u>determinant</u> of a 3 x 3 matrix. Logical since it being 0 vs. not is what determines invertibility.

But importantly, we see that this process created many 2 x 2 determinants as entries along the way. That association is very important in formalizing the concept of determinant to any size square matrix.

For 2 x 2 would play out this way, for a nonzero entry:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \sim \begin{bmatrix} a & b \\ ac & ad \end{bmatrix} \sim \begin{bmatrix} a & b \\ 0 & ad - bc \end{bmatrix}$$

The determinant we are acccustommed to!

So formalizing this for 3 x 3

$$A = \begin{bmatrix} \boxed{a_{11} \quad a_{12} \quad a_{13}} \\ a_{21} \quad a_{22} \quad a_{23} \\ a_{31} \quad a_{32} \quad a_{33} \end{bmatrix}$$

Let's "expand" in first row, as boxed. Each entry, multiply it by the determinant of the matrix obtained by deleting the row and column through that entry, alternating signs of terms. This gives

$$\det A = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Absolute value bars mean determinant in this context.

Why this works a bit more involved, but in our computation turning A into upper triangular, if we check what the bottom right entry would have become it would be a lot like this so logical from that perspective.

One very nice thing about this is that we can "expand" in any row or column and get same value, that sort of validates importance of determinant, it's not just about determining if matrix is invertible or not.

$$A = \begin{bmatrix} a_{11} & \boxed{a_{12}} & a_{13} \\ a_{21} & \boxed{a_{22}} & a_{23} \\ a_{31} & \boxed{a_{32}} & a_{33} \end{bmatrix}$$

Let us expand in second column instead.

$$\det A = -a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{22}\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{32}\begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$$

If we continue with this and earlier one, we will see they are the same.

This being the case, when finding determinant of a reasonably large matrix, expanding in a row or column with as many 0s as possible is a natural strategy.

Before thinking about that, the concept of removing the row and column through a particular entry is a common idea in this context so is useful to give that formal terminology.

For a square matrix A whose ij-th entry (i-th row, j-th column) is as usual denoted a_{ij}, let A_{ij} denote the matrix where we remove the i-th row and j-th column from A. As alluded, the determinant of this matrix is relevant so we create terminology related to it.

The <u>cofactor</u> of entry a_{ij}, denoted C_{ij}, is given by

$$C_{ij} = (-1)^{i+j} \cdot \det A_{ij}$$

That is, cofactor of an entry is the determinant of the matrix created by removing the row and column through that entry, with potentially a sign switch. The sign switch alternates by entry.

Under that notation, the determinant of A, via expanding in the p-th row is

$$\det A = \sum_{j=1}^{n} a_{pj} C_{pj}$$

or in the q-th column

$$\det A = \sum_{i=1}^{n} a_{iq} C_{iq}$$

3.1.13

$$\begin{vmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & -2 & 3 \\ 0 & 0 & 9 & -1 & 2 \end{vmatrix}$$

Expand in second row since four of five entries are 0:

$$= \quad -2 \begin{vmatrix} 4 & 0 & 3 & -5 \\ 7 & 3 & 4 & 8 \\ 5 & 0 & -2 & 3 \\ 0 & 0 & -1 & 2 \end{vmatrix}$$

Now second column since that has all but one entry 0:

$$= \quad -2(3) \begin{vmatrix} 4 & 3 & -5 \\ 5 & -2 & 3 \\ 0 & -1 & 2 \end{vmatrix}$$

Only one zero entry left so expand in the row or column which includes that:

$$= \quad -6 \left[4 \begin{vmatrix} -2 & 3 \\ -1 & 2 \end{vmatrix} - 5 \begin{vmatrix} 3 & -5 \\ -1 & 2 \end{vmatrix} \right]$$

$= \quad -6[4(-4 + 3) - 5(6 - 5)]$

$= \quad -6[-4 - 5] = 54$

What if we wanted to find determinant of a 5 x 5 matrix which had no 0 entries?

Probably such a question wouldn't be practically reasonable unless there was some trick. But how would you try it if forced?

There are no 0 entries, but what if you created some via row operations?

Reasonable thought, after all row operations do make many things easier. But do row operations change the determinant?

Row operations after all don't change the solution to the homogeneous equation when thinking of the matrix as a coefficient matrix, but on the other hand they do potentially change the span of the columns.

In short they do change some things but not others. So what about determinant? And there are three different types of row operations, might each have a different effect?

Have to investigate! That's what next section is about.